Session 16 · Tuesday 20 October 2026 · Three to work, one worked for you, one stretch. Everything starts at zero, and that is the whole difficulty.

Ticks, glitches and depth

The rule for every table on this sheet. Time runs in ticks. At each tick every gene looks at the values of its inputs **as they were at the previous tick** and updates. All genes update together. At $t = 0$ the cell has just been made, so **every gene is at 0** and the inputs are already at their given values.

The three kinds of glitch, named by what the output was supposed to be doing:

	THE OUTPUT SHOULD HAVE	WHAT IT DOES INSTEAD
static-0	stayed at 0	goes up and comes back
static-1	stayed at 1	goes down and comes back
dynamic	changed once	changes more than once on the way

Two of these can happen in a cell that has just been made. The third cannot, and item 3(d) is where you find out why.

One number for depth. A chain of n genes, each driving the next, with a step at the input and every protein sharing a lifetime τ:

$$t_{1/2}(n) \;=\; \text{time for the output to reach half its final value,} \qquad t_{1/2}(n) \;\approx\; \left(n - \tfrac{1}{3}\right)\tau.$$

Two promoters, and the difference between them. A gene behind a plain repressed promoter is ON unless a repressor is bound. A gene behind a promoter that **requires an activator** is OFF unless that activator is present; we write $\mathrm{BUF}(X)$ for the one-input version, a gene that comes on one tick after X appears and is off before that.

1 — Fully worked: the classifier, $A = 1$, $B = 0$

The circuit from class. $R_1 = \text{NOT } B$. R_2 is one gene behind two promoters, one repressed by A and one by R_1, so $R_2 = \text{NAND}(A, R_1)$. $R_3 = \text{NOT } R_2$, and R_3 is the output. With $A = 1$ and $B = 0$ the table settles to $R_3 = 1$.

t	R_1	R_2	R_3
0	0	0	0
1	1	1	1
2	1	0	0
3	1	0	1
4	1	0	1

Read the last column: 0, 1, 0, 1, 1. The output was supposed to change once, from its reset value to 1. It changed three times. **That is a dynamic hazard**, and it is the case whose correct answer is 1 — the circuit arrives at the right answer having first given it, taken it back, and given it again.

Why does that step follow? — at $t = 1$ every gene whose repressors are all still absent turns on at once. R_2 is a NAND, ON unless both its repressors are present, and at $t = 0$ neither is. R_3 is a NOT, ON unless its repressor is present, and at $t = 0$ R_2 is not. A gene reading only *other genes* is therefore switched on in an empty cell. A gene reading an input directly is not: $R_1 = \text{NOT } B$ with $B = 1$ would stay off.

2 — Compile XNOR, and count the depths

XNOR is 1 when A and B agree. Build it from NOT and NOR gates only.

The recipe, if you want one. Write the output as an OR over the rows where it is 1; turn each row into a NOR of the complements of its literals; combine the row terms with a NOR; invert. You may use any other route that works.

Write the network as a list of genes, each line saying what represses what. You will not need all these lines.

Count gates along every route from an input to the output. Count the output gene itself.

$$\text{longest path} = _____ \text{ genes}, \qquad \text{shortest path} = _____ \text{ genes}$$

Why does that step follow? — a different group will hand in a different network that computes the same table. What, if anything, makes one of them better?

3 — A different circuit

Three genes, two inputs, wired differently from item 1:

$$R_1 = \text{NOT } A, \qquad R_2 = \text{NOR}(B,\, R_1), \qquad \text{out} = \text{NOT } R_2.$$

Fill five rows, $t = 0$ to $t = 4$, for each input case. Every gene starts at 0.

t	R_1	R_2	OUT		t	R_1	R_2	OUT
$A = 0,\ B = 0$					$A = 0,\ B = 1$			
0					0			
1					1			
2					2			
3					3			
4					4			
$A = 1,\ B = 0$					$A = 1,\ B = 1$			
0					0			
1					1			
2					2			
3					3			
4					4			

(a) What function does it settle to? Write the truth table.

A	B	out
0	0	__
0	1	__
1	0	__
1	1	__

(b) Which input cases glitch, and which kind is each? One line per case.

CASE	GLITCHES?	WHICH KIND
$A = 0,\ B = 0$		
$A = 0,\ B = 1$		
$A = 1,\ B = 0$		
$A = 1,\ B = 1$		

(c) Look at the $t = 1$ row of all four of your tables. What do the four output values have in common, and which gene decided that?

(d) Now stop starting from an empty cell. Let the circuit settle at $A = 1,\ B = 1$ — you have that state in your fourth table — and then drop **both** inputs to 0 at $t = 0$, keeping the genes where they settled.

t	R_1	R_2	OUT
0			
1			
2			
3			
4			

The output is 1 before the change and 1 after it. Which kind of glitch is this, and why could it not have appeared anywhere in part (b)?

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4 — How late is a four-gene chain?

Four genes in a chain, every protein with the same lifetime τ, a step at the input at $t = 0$. **Use the rule of thumb, not the exact sum.**

$$t_{1/2}(4) \approx _____\ \tau$$

By the rule of thumb, how much later is that than the three-gene chain?

$$_____\ \tau$$

Why does that step follow? — would you have guessed that number before this session? Most people underestimate depth. What about the **first** stage makes it misleading?

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5 — Stretch: what actually decides it

A circuit is **fail-safe** if its output gene is still off one tick after the cell is made — that is, if the output promoter reads 0 while every protein in the circuit is still absent. A NOT or NOR output is never fail-safe. A circuit is **balanced** if every route from any input to the output has the same number of genes.

Here are three circuits. All three compute something; two of them glitch.

	THE GENES	OFF WHEN EMPTY?	PATHS EQUAL?
(i)	item 1, then $\mathbf{Act} = \mathbf{NOT}\, R_2$, $R_3 = \mathbf{BUF}(\mathbf{Act})$	yes	no
(ii)	$\neg A$, $\neg B$, $Z = \mathbf{NOR}(\neg A, \neg B)$, $\mathbf{out} = \mathbf{BUF}(Z)$	yes	**yes**
(iii)	$g = \mathbf{BUF}(A)$, $\mathbf{out} =$ a promoter needing g, blocked by B	yes	no

Run three ticks of (ii) and three of (iii), at $A = 0,\ B = 0$.

(a) (i) and (ii) glitch; (iii) does not. So neither "OFF when empty" nor "equal paths" is the thing that matters. **What is true of (iii) that is false of both the others?**

(b) State it as a rule about *when a signal arrives at a promoter*, and say which kind of site — activating or repressing — you are allowed to be late at.